# KNITTRACK TIRUPPUR — TECHNICAL DESIGN DOCUMENT

Production Order Tracking System with Multi-Tenant Buyer View & AI Engine

Client: Company A, Tiruppur | Author: Senior Full-Stack Engineer | Date: November 2026

**VERIFIED PROJECT REPOSITORIES & ENDPOINTS**

GitHub Repository: https://github.com/Subha23456/Knitwear-exporter-s-production-order-tracking-app-with-buyer-view/
Live Application URL: https://ais-pre-a2bkzzm4scpewhcprbh3ck-235016284191.asia-southeast1.run.app
Development Preview: https://ais-dev-a2bkzzm4scpewhcprbh3ck-235016284191.asia-southeast1.run.app

## 1. Executive Summary & The Client Problem

Company A is a leading knitwear exporter based in Tiruppur, Tamil Nadu, supplying knitwear garments to top UK and EU retailers (annual turnover Rs 95–110 Crore across 500 export orders/year). The company operates 3 owned manufacturing units (Unit 1 Kangeyam Rd, Unit 2 Angeripalayam, Unit 3 Netaji Apparel Park) along with contracted printing and washing facilities, employing ~900 workers and an office team of 35 merchandisers.

Today, orders are tracked manually through fragmented Excel workbooks and phone calls. This operational blindness leads to Rs 1.08 Crore in annual leakage:

- 8% of shipments miss ex-factory deadlines (40 orders per year).
- 15 emergency air shipments incurred at Rs 3.50 Lakh penalty each = Rs 52.50 Lakh.
- 25 late sea shipments hit by a 5% buyer discount penalty (Rs 1.05L average) = Rs 26.25 Lakh.
- 12 merchandisers spend 2 hours daily chasing floor supervisors = 7,200 hours @ Rs 400/hr = Rs 28.80 Lakh.
- Total Annual Financial Loss: Rs 1.0755 Crore (~Rs 1.08 Crore).

## 2. System Architecture & Technology Stack

The application is architected as a modular, full-stack, single-tenant/multi-persona web application. The frontend communicates with a secure Node.js/Express backend that interfaces with SQLite via better-sqlite3 and the Google Gemini 3.8 Flash model via @google/genai.

- Frontend: React 19, TypeScript 5.7, Tailwind CSS, Lucide Icons, Vite.
- Backend: Node.js 22, Express 4.21, TypeScript native type stripping, WAL SQLite.
- Database: SQLite embedded database (data/knittrack.db) with foreign keys and WAL mode.
- AI & NLP: Google Gemini 3.8 Flash API with structured JSON output and deterministic Tamil NLP fallback.
- Testing: Vitest test framework with 24 automated unit and integration test cases.

## 3. Database Schema & Data Model (SQLite)

Data persistence is managed via an embedded SQLite database configured with foreign keys and Write-Ahead Logging (WAL) for concurrent reads:

- users: id (PK), name, role (merchandiser|supervisor|buyer|md), email (UNIQUE), password, organization, unit, buyerId, preferredLanguage.
- sessions: token (PK, 64-char crypto hex), userId (FK -> users.id), createdAt, expiresAt.
- orders: id (PK), poNumber (UNIQUE), buyerId, buyerName, styleDescription, productType, garmentImageUrl, unit, totalQuantity, totalQuantityDone, fobPricePerPieceINR, orderValueINR, orderDate, exFactoryDate, currentStage, status, merchandiserId, hasUnconfirmedDelayWarning, confirmedBuyerWarningJson, colorsJson.
- milestones: id (PK), orderId (FK -> orders.id ON DELETE CASCADE), stage, targetStartDate, targetEndDate, actualStartDate, actualEndDate, targetQuantity, completedQuantity, status, responsibleUnit.
- daily_logs: id (PK), orderId (FK -> orders.id ON DELETE CASCADE), date, stage, quantityDoneToday, supervisorId, unit, photoUrl, originalNoteTamil, englishTranslation, aiReasonCategory, aiConfidence, merchandiserReviewStatus, isSharedWithBuyer, createdAt.
- date_change_audits: id (PK), orderId (FK -> orders.id ON DELETE CASCADE), fieldChanged, previousValue, newValue, reason, authorizedBy, timestamp.

## 4. Run-Rate Delay Mathematical Formulas & Exact Rules

The run-rate calculation engine (src/utils/runRate.ts) computes real-time production velocity without hardcoded results:

- Remaining Quantity (R) = Total Quantity - Completed Quantity.
- Working Days Remaining (D) = Working days between evaluation date and ex-factory date, excluding Sundays.
- Required Daily Production = R / D (in pieces per working day).
- Actual Daily Production = Average completed quantity from recent floor logs (last 3 entries).
- Projected Completion Days = R / Actual Daily Production.
- Projected Completion Date = Today + Projected Completion Days in working days.
- Projected Slip Days = Calendar difference between Projected Date and Ex-Factory Date.

### Status Decision Rules:

- Green Status: Actual daily output meets or exceeds required daily output (Ratio >= 1.0) or R = 0.
- Amber Status: Actual daily output is 80% to less than 100% of required daily output (0.80 <= Ratio < 1.0).
- Red Status: Actual output is below 80% of required output (Ratio < 0.80) or ex-factory date has passed with unfinished balance.
- Financial Exposure: Red status triggers Rs 3,50,000 air-freight surcharge risk + 5% buyer late discount.

## 5. Floor Supervisor Mobile & Tamil AI Engine

Designed for factory floor ergonomics and low-cost Android phones (2–3 GB RAM). Includes quick touch steppers (+100, +250, +500), milestone stage selector, and live bilingual toggle (Tamil / English).

- Web Speech API Audio Recognition: Captures voice input directly in Tamil (ta-IN) and English (en-IN).
- Offline Storage Queue: Unsent logs store in localStorage with pending counter and 1-tap re-sync.
- Gemini 3.8 Flash AI Model: Analyzes Tamil notes and outputs English translation, root-cause category, and severity.
- 8 Standardized Categories: Yarn & Fabric Delay, Machine Breakdown, Needle Breakage / QC Alert, Printing Screen Defect, Subcontract Processor Delay, Absenteeism / Power Outage, Trims Shortage, Normal Production.
- Deterministic Fallback: Built-in offline keyword dictionary ensures 100% classification uptime even if Gemini API key is unset.
- Human-in-the-Loop Quarantine: Supervisor delay warnings remain quarantined until approved by a merchandiser.

## 6. Security & Multi-Tenant Buyer Isolation

Strict server-side tenant isolation guarantees that Buyer B (UK) cannot see Buyer C (Germany) orders, styles, or photos:

- Session-Based Auth: Authenticated Bearer tokens checked via requireAuth and requireRole middlewares.
- Zero Trust Query Parameters: req.user.buyerId is derived exclusively from the session; buyer query params are discarded.
- Cross-Tenant Defense: Accessing another buyer's order ID returns HTTP 404 (tested in test suite).
- UI Lock: When a buyer signs in, role switcher dropdowns and staff navigation links are completely removed.
- Export Isolation: /api/export/buyer-report scopes CSV data strictly to the requesting buyer.

## 7. Automated Verification & Test Suite

The test suite is built on Vitest and runs via "npm test". All 24 tests pass cleanly in 633ms:

- src/test/runRate.test.ts (8 tests): Working days, Sundays exclusion, Green, Amber, Red thresholds, zero working days, completed orders, zero production logs.
- src/test/tamilClassification.test.ts (8 tests): Machine breakdown, yarn delay, screen defect, needle breakage, subcontract delay, power outage, note preservation.
- src/test/securityAndIsolation.test.ts (8 tests): Buyer B vs Buyer C isolation, cross-tenant 404 block, merchandiser visibility, immutable date audits, warning quarantine.

## 8. Reviewer Demo Logins

- Priya Rajan (Merchandiser): priya.rajan@companya-knits.com | Password: tiruppur123 | Opens Production Board
- Murugan Kandasamy (Supervisor): murugan.unit1@companya-knits.com | Password: tiruppur123 | Opens Mobile View

(Tamil)

- John Smith (Buyer - UK): j.smith@companyb-retail.co.uk | Password: tiruppur123 | Opens Buyer Portal (B-PO only)
- Hans Weber (Buyer - Germany): h.weber@companyc-textil.de | Password: tiruppur123 | Opens Buyer Portal (C-PO only)
- K. Shanmugam (MD): ks@companya-knits.com | Password: tiruppur123 | Opens MD Financial Dashboard